

NPACN Lab Eval Cheatsheet

Cisco Packet Tracer — RIP, OSPF, DHCP, NAT, Subnetting

Last updated: April 8, 2026

0. Quick Reference — The Order of Operations

Always follow this order:

1. Draw the topology on paper, label ALL interfaces and IPs
2. Assign IPs to every router interface (+ **no shutdown**)
3. **clock rate** on DCE side of serial links
4. Configure routing (RIP / OSPF)
5. Configure DHCP pools (if needed)
6. Configure NAT (if needed)
7. Verify with **ping** and **show** commands

1. IP Addressing & Subnetting (The #1 Thing That Bricks You)

1.1 The Golden Rules

- Each **link** (wire) between two devices must be its own **subnet**
- Two ends of the same wire = **same network, different host IPs**
- Gateway IP on a PC = the router interface IP on that same subnet
- **no shutdown** on EVERY interface or nothing works

1.2 Common Subnet Masks

CIDR	Subnet Mask	Hosts	Use Case
/8	255.0.0.0	16M	Class A
/16	255.255.0.0	65K	Class B
/24	255.255.255.0	254	Most common in labs
/25	255.255.255.128	126	Split a /24 in half
/26	255.255.255.192	62	Quarter of /24
/27	255.255.255.224	30	
/28	255.255.255.240	14	
/30	255.255.255.252	2	Point-to-point serial links

1.3 How to Plan IPs for a Topology

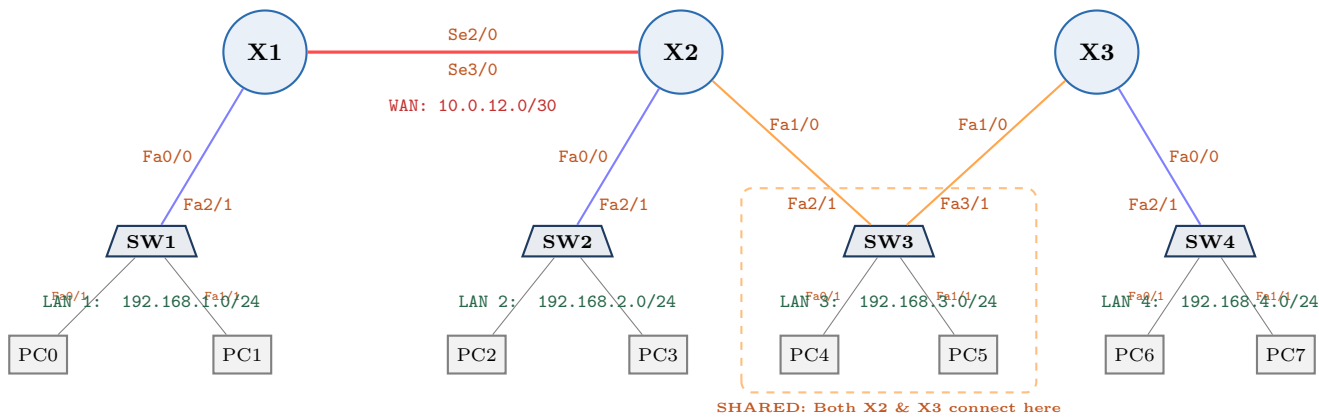
Strategy: Use 10.x.x.x or 192.168.x.x networks.

For each link, pick a different third octet:

- Router1–Router2 serial link: 10.0.12.0/30 → R1 gets .1, R2 gets .2
- Router2–Router3 serial link: 10.0.23.0/30 → R2 gets .1, R3 gets .2
- Router1 LAN (to switch/PCs): 192.168.1.0/24 → R1 Fa0/0 = .1, PCs = .2+
- Router2 LAN: 192.168.2.0/24 → R2 Fa0/0 = .1
- Router3 LAN: 192.168.3.0/24 → R3 Fa0/0 = .1

The trick: **each wire = unique subnet**. Never reuse the same subnet on different links!

1.4 Reference Topology (From Your Exam Paper)



1.5 IP Address Plan for This Topology

Device	Interface	IP Address	Network / Role
X1	Fa0/0	192.168.1.1 / 255.255.255.0	LAN 1 (gateway for PC0, PC1)
	Se2/0	10.0.12.1 / 255.255.255.252	WAN serial link (DCE)
X2	Se3/0	10.0.12.2 / 255.255.255.252	WAN serial link (DTE)
	Fa0/0	192.168.2.1 / 255.255.255.0	LAN 2 (gateway for PC2, PC3)
	Fa1/0	192.168.3.1 / 255.255.255.0	LAN 3 shared (gateway for PC4, PC5)
X3	Fa1/0	192.168.3.2 / 255.255.255.0	LAN 3 shared (same subnet as X2 Fa1/0)
	Fa0/0	192.168.4.1 / 255.255.255.0	LAN 4 (gateway for PC6, PC7)
PC0, PC1	Fa0	192.168.1.x / 255.255.255.0	Gateway: 192.168.1.1
PC2, PC3	Fa0	192.168.2.x / 255.255.255.0	Gateway: 192.168.2.1
PC4, PC5	Fa0	192.168.3.x / 255.255.255.0	Gateway: 192.168.3.1
PC6, PC7	Fa0	192.168.4.x / 255.255.255.0	Gateway: 192.168.4.1

LAN 3 is SHARED between X2 and X3! Both routers have an interface (Fa1/0) plugged into the **same switch (SW3)**. So X2's Fa1/0 and X3's Fa1/0 are in the **same 192.168.3.0/24 subnet**. This means X2 and X3 can reach each other directly through LAN 3 — no serial cable needed between them.

PC4 and PC5 should use **192.168.3.1 (X2)** as their gateway. X3 also sits on this LAN at 192.168.3.2.

1.6 Step-by-Step Configuration (RIP)

STEP 1: Configure Router X1

X1 — Complete Configuration

```
enable
configure terminal
hostname X1

! ---- LAN 1 interface (Fa0/0 -> SW1 -> PC0, PC1) ----
interface FastEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

! ---- WAN Serial link to X2 (DCE side) ----
interface Serial 2/0
ip address 10.0.12.1 255.255.255.252
clock rate 64000
no shutdown
exit
```

```
! ---- RIP routing ----
router rip
  version 2
  no auto-summary
  network 192.168.1.0
  network 10.0.12.0
  exit

end
copy running-config startup-config
```

STEP 2: Configure Router X2

X2 — Complete Configuration (3 interfaces!)

```
enable
configure terminal
hostname X2

! ---- WAN Serial link to X1 (DTE side, NO clock rate) ----
interface Serial 3/0
  ip address 10.0.12.2 255.255.255.252
  no shutdown
  exit

! ---- LAN 2 interface (Fa0/0 -> SW2 -> PC2, PC3) ----
interface FastEthernet 0/0
  ip address 192.168.2.1 255.255.255.0
  no shutdown
  exit

! ---- LAN 3 SHARED interface (Fa1/0 -> SW3 -> PC4, PC5) ----
interface FastEthernet 1/0
  ip address 192.168.3.1 255.255.255.0
  no shutdown
  exit

! ---- RIP routing (advertise ALL 3 connected networks) ----
router rip
  version 2
  no auto-summary
  network 10.0.12.0
  network 192.168.2.0
  network 192.168.3.0
  exit

end
copy running-config startup-config
```

STEP 3: Configure Router X3

X3 — Complete Configuration

```
enable
configure terminal
hostname X3

! ---- LAN 3 SHARED interface (Fa1/0 -> SW3 -> PC4, PC5) ----
! Same subnet as X2's Fa1/0, but different host IP (.2)
interface FastEthernet 1/0
  ip address 192.168.3.2 255.255.255.0
  no shutdown
  exit

! ---- LAN 4 interface (Fa0/0 -> SW4 -> PC6, PC7) ----
interface FastEthernet 0/0
  ip address 192.168.4.1 255.255.255.0
  no shutdown
  exit
```

```
! ---- RIP routing ----
router rip
  version 2
  no auto-summary
  network 192.168.3.0
  network 192.168.4.0
  exit

end
copy running-config startup-config
```

STEP 4: Configure PCs

PC	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.100	255.255.255.0	192.168.1.1
PC1	192.168.1.101	255.255.255.0	192.168.1.1
PC2	192.168.2.100	255.255.255.0	192.168.2.1
PC3	192.168.2.101	255.255.255.0	192.168.2.1
PC4	192.168.3.100	255.255.255.0	192.168.3.1
PC5	192.168.3.101	255.255.255.0	192.168.3.1
PC6	192.168.4.100	255.255.255.0	192.168.4.1
PC7	192.168.4.101	255.255.255.0	192.168.4.1

STEP 5: Verify

```
! On X1:
X1# show ip route
! C 192.168.1.0/24 (Fa0/0 - directly connected)
! C 10.0.12.0/30 (Se2/0 - directly connected)
! R 192.168.2.0/24 via 10.0.12.2 (learned via RIP from X2)
! R 192.168.3.0/24 via 10.0.12.2 (learned via RIP from X2)
! R 192.168.4.0/24 via 10.0.12.2 (learned via RIP from X2->X3)

! On X2:
X2# show ip route
! C 10.0.12.0/30 (Se3/0)
! C 192.168.2.0/24 (Fa0/0)
! C 192.168.3.0/24 (Fa1/0)
! R 192.168.1.0/24 via 10.0.12.1 (from X1)
! R 192.168.4.0/24 via 192.168.3.2 (from X3 through LAN 3!)

! On X3:
X3# show ip route
! C 192.168.3.0/24 (Fa1/0)
! C 192.168.4.0/24 (Fa0/0)
! R 192.168.2.0/24 via 192.168.3.1 (from X2 through LAN 3)
! R 192.168.1.0/24 via 192.168.3.1 (from X2)
! R 10.0.12.0/30 via 192.168.3.1 (from X2)

! Test pings:
! PC0 -> PC7 (should work: PC0->X1->X2->X3->PC7)
! PC4 -> PC6 (should work: PC4->X3->PC6, or PC4->X2->X3->PC6)
```

How RIP propagates routes here:

1. X1 tells X2 about 192.168.1.0 through the serial link
 2. X2 tells X1 about 192.168.2.0 and 192.168.3.0 through the serial link
 3. X2 tells X3 about 192.168.1.0, 192.168.2.0, 10.0.12.0 through **LAN 3** (SW3)
 4. X3 tells X2 about 192.168.4.0 through **LAN 3**
 5. X2 then forwards 192.168.4.0 to X1 through the serial link
- X2 is the “bridge” between all networks. X3 reaches X1 through X2 via LAN 3.

2. Basic Router & Interface Configuration

2.1 Entering Configuration Mode

```
Router> enable                ! User -> Privileged mode
Router# configure terminal     ! Privileged -> Global config
Router(config)# hostname R1    ! Set hostname
R1(config)#
```

2.2 Configuring a FastEthernet Interface (LAN side)

```
R1(config)# interface FastEthernet 0/0    ! or Fa0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown               ! CRITICAL - turn it ON
R1(config-if)# exit
```

2.3 Configuring a Serial Interface (WAN / Router-to-Router)

```
R1(config)# interface Serial 2/0          ! or Se2/0
R1(config-if)# ip address 10.0.12.1 255.255.255.252 ! /30 mask
R1(config-if)# clock rate 64000          ! ONLY on DCE side
R1(config-if)# no shutdown
R1(config-if)# exit
```

clock rate: Only set this on the **DCE** end of the serial cable.

To check which end you are: `show controllers serial 2/0` — look for “DCE” or “DTE”.

If you’re on the DCE side, you **must** set clock rate. If DTE, don’t set it.

Common values: 64000, 128000, 256000

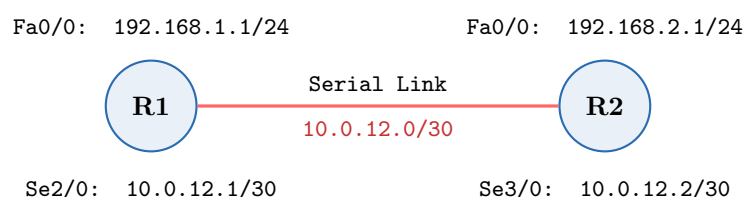
2.4 Configuring a GigabitEthernet Interface

```
R1(config)# interface GigabitEthernet 0/0 ! or Gig0/0 or g0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

2.5 Saving Configuration

```
R1# copy running-config startup-config
! or shortcut:
R1# wr
```

3. RIP — Connecting 2+ Routers



3.1 Step-by-Step: RIP between R1 and R2

Step 1: Assign IPs on both routers (see Section 2)

Step 2: Configure RIP on R1

```
R1(config)# router rip
R1(config-router)# version 2           ! Always use v2
R1(config-router)# no auto-summary     ! Prevent classful chaos
R1(config-router)# network 192.168.1.0 ! R1's LAN
R1(config-router)# network 10.0.12.0   ! The serial link subnet
R1(config-router)# exit
```

Step 3: Configure RIP on R2

```
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 192.168.2.0 ! R2's LAN
R2(config-router)# network 10.0.12.0   ! Same serial link subnet
R2(config-router)# exit
```

RIP uses CLASSFUL network addresses!

Under **network**, you provide the **major network** (no subnet mask).

network 10.0.12.0 → RIP advertises everything in 10.x.x.x unless **no auto-summary**.

ALWAYS add: version 2 + no auto-summary

Without these, RIP silently fails across subnets.

3.2 RIP with 3 Routers

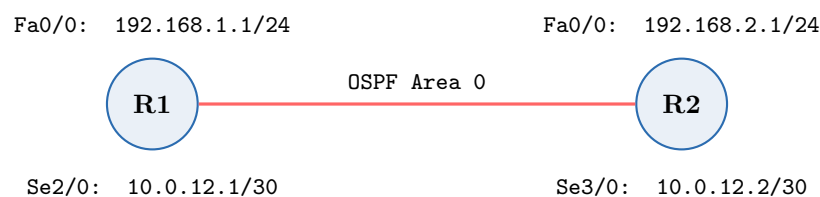
Just add every directly-connected network on each router:

```
! -- R2 (the middle router) --
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 10.0.12.0   ! Link to R1
R2(config-router)# network 10.0.23.0   ! Link to R3
R2(config-router)# network 192.168.2.0 ! R2's own LAN
R2(config-router)# exit
```

3.3 Verify RIP

```
R1# show ip route           ! Look for "R" entries (RIP routes)
R1# show ip rip database    ! RIP-specific info
R1# show ip protocols       ! Which routing protocol is running
```

4. OSPF — Connecting 2+ Routers



4.1 Step-by-Step: OSPF between R1 and R2

Step 1: Assign IPs on both routers (see Section 2)

Step 2: Configure OSPF on R1

```
R1(config)# router ospf 1                ! 1 = process ID (local)
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.0.12.0 0.0.0.3 area 0
R1(config-router)# exit
```

Step 3: Configure OSPF on R2

```
R2(config)# router ospf 1
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# network 10.0.12.0 0.0.0.3 area 0
R2(config-router)# exit
```

OSPF uses WILDCARD masks, NOT subnet masks!

Wildcard = 255.255.255.255 minus subnet mask.

Subnet mask 255.255.255.0 → Wildcard = **0.0.0.255**
 Subnet mask 255.255.255.252 (/30) → Wildcard = **0.0.0.3**
 Subnet mask 255.255.255.240 (/28) → Wildcard = **0.0.0.15**
 Subnet mask 255.255.0.0 → Wildcard = **0.0.255.255**

Process ID (the “1”): Can be any number 1–65535. It’s LOCAL to the router.
 Does **not** need to match between routers (but often kept same for clarity).

Area: Use area 0 for single-area OSPF (almost always in labs).

4.2 Wildcard Mask Quick-Calc Cheat

/CIDR	Subnet Mask	Wildcard Mask
/24	255.255.255.0	0.0.0.255
/25	255.255.255.128	0.0.0.127
/26	255.255.255.192	0.0.0.63
/27	255.255.255.224	0.0.0.31
/28	255.255.255.240	0.0.0.15
/30	255.255.255.252	0.0.0.3
/16	255.255.0.0	0.0.255.255
/8	255.0.0.0	0.255.255.255

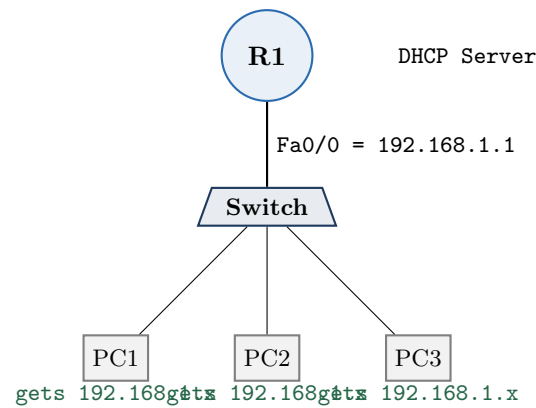
4.3 OSPF with 3 Routers

```
! -- R2 (middle router) --
R2(config)# router ospf 1
R2(config-router)# network 10.0.12.0 0.0.0.3 area 0      ! Link to R1
R2(config-router)# network 10.0.23.0 0.0.0.3 area 0      ! Link to R3
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0  ! R2 LAN
R2(config-router)# exit
```

4.4 Verify OSPF

```
R1# show ip ospf neighbor      ! Must see neighbors (FULL state)
R1# show ip route              ! Look for "O" entries (OSPF routes)
R1# show ip ospf interface     ! OSPF details per interface
R1# show ip protocols          ! Verify OSPF is running
```

5. DHCP — Router Assigns IPs to PCs Automatically



5.1 Configure DHCP on the Router

```

! Step 1: Exclude the router's own IP (and any static IPs)
R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10

! Step 2: Create a DHCP pool
R1(config)# ip dhcp pool LAN_POOL
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.1.1      ! Gateway for PCs
R1(dhcp-config)# dns-server 8.8.8.8              ! Optional
R1(dhcp-config)# exit
  
```

5.2 Configure PCs to Use DHCP

In Packet Tracer:

1. Click on the PC → **Desktop** tab → **IP Configuration**
2. Select **DHCP** (instead of Static)
3. Wait a moment — IP, mask, gateway should auto-fill

If nothing fills in → check that the router interface has an IP + no shutdown!

5.3 DHCP for Multiple LANs

Create one pool per LAN subnet:

```

! Pool for LAN 1 (Fa0/0 side)
R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
R1(config)# ip dhcp pool P00L1
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.1.1
R1(dhcp-config)# exit

! Pool for LAN 2 (Fa0/1 side)
R1(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.10
R1(config)# ip dhcp pool P00L2
R1(dhcp-config)# network 192.168.2.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.2.1
R1(dhcp-config)# exit
  
```

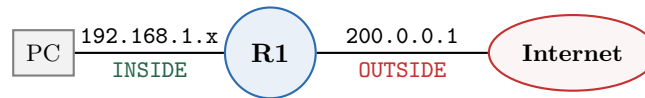
5.4 Verify DHCP

```

R1# show ip dhcp binding      ! See which IPs were given out
R1# show ip dhcp pool         ! Pool statistics
  
```

```
R1# show ip dhcp conflict      ! Any IP conflicts
```

6. NAT — Network Address Translation



NAT translates private ↔ public

6.1 NAT Terminology

Term	Meaning
Inside Local	Private IP of host (e.g., 192.168.1.10)
Inside Global	Public IP seen on Internet (e.g., 200.0.0.1)
Outside Local	IP of external host as seen from inside
Outside Global	Actual IP of external host
ip nat inside	Put this on the LAN-facing interface
ip nat outside	Put this on the WAN/Internet-facing interface

6.2 Static NAT (1-to-1 mapping)

```

! Map private 192.168.1.10 to public 200.0.0.10 (one host)
R1(config)# ip nat inside source static 192.168.1.10 200.0.0.10

! Mark interfaces
R1(config)# interface Fa0/0
R1(config-if)# ip nat inside
R1(config-if)# exit

R1(config)# interface Se2/0      ! or whatever faces "outside"
R1(config-if)# ip nat outside
R1(config-if)# exit

```

6.3 Dynamic NAT (Pool of public IPs)

```

! Step 1: Define the public IP pool
R1(config)# ip nat pool MY_POOL 200.0.0.1 200.0.0.10 netmask 255.255.255.0

! Step 2: ACL to match which private IPs get translated
R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255

! Step 3: Link ACL to pool
R1(config)# ip nat inside source list 1 pool MY_POOL

! Step 4: Mark interfaces
R1(config)# interface Fa0/0
R1(config-if)# ip nat inside
R1(config-if)# exit
R1(config)# interface Se2/0
R1(config-if)# ip nat outside
R1(config-if)# exit

```

6.4 PAT / NAT Overload (Most Common — Many-to-1)

```

! This is what you'll use 90% of the time in labs
! All internal hosts share ONE public IP

```

```

! Step 1: ACL to match internal network
R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255

! Step 2: NAT overload using the outside interface's IP
R1(config)# ip nat inside source list 1 interface Se2/0 overload

! Step 3: Mark interfaces
R1(config)# interface Fa0/0
R1(config-if)# ip nat inside
R1(config-if)# exit
R1(config)# interface Se2/0
R1(config-if)# ip nat outside
R1(config-if)# exit

```

PAT/Overload is the most common. Use this unless told otherwise.
The keyword **overload** is what makes it PAT instead of regular dynamic NAT.

6.5 Verify NAT

```

R1# show ip nat translations      ! Active translations
R1# show ip nat statistics       ! Hit counts
R1# clear ip nat translation *    ! Clear all translations

```

7. Static Routing (Bonus — Sometimes Asked)

```

! Syntax: ip route [dest-network] [mask] [next-hop-IP or exit-interface]

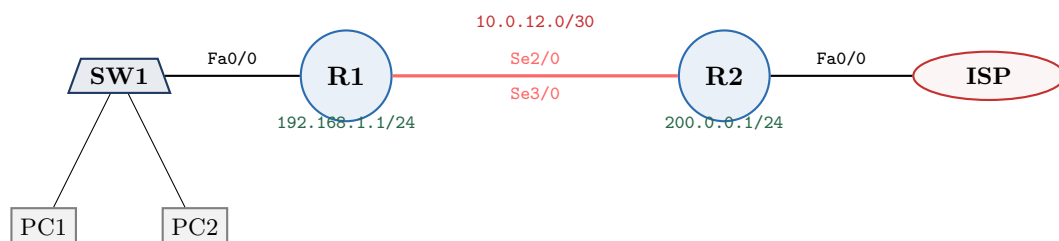
! On R1: to reach R2's LAN (192.168.2.0/24), go via 10.0.12.2
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.12.2

! On R2: to reach R1's LAN (192.168.1.0/24), go via 10.0.12.1
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.12.1

! Default route (send everything unknown to a next-hop)
R1(config)# ip route 0.0.0.0 0.0.0.0 10.0.12.2

```

8. Complete Worked Example: 2 Routers + RIP + DHCP + NAT



R1 — Full Config

```

enable
configure terminal
hostname R1

! -- Interfaces --
interface Fa0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown

```

```
exit

interface Se2/0
 ip address 10.0.12.1 255.255.255.252
 clock rate 64000
 ip nat outside
 no shutdown
 exit

! -- RIP --
router rip
 version 2
 no auto-summary
 network 192.168.1.0
 network 10.0.12.0
 exit

! -- DHCP --
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp pool LAN_POOL
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
 exit

! -- NAT (PAT) --
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface Se2/0 overload

end
copy running-config startup-config
```

R2 — Full Config

```
enable
configure terminal
hostname R2

interface Se3/0
 ip address 10.0.12.2 255.255.255.252
 no shutdown
 exit

interface Fa0/0
 ip address 200.0.0.1 255.255.255.0
 no shutdown
 exit

router rip
 version 2
 no auto-summary
 network 10.0.12.0
 network 200.0.0.0
 exit

end
copy running-config startup-config
```

9. Complete Worked Example: 2 Routers + OSPF + DHCP

R1 with OSPF + DHCP

```
enable
configure terminal
hostname R1
```

```
interface Fa0/0
  ip address 192.168.1.1 255.255.255.0
  no shutdown
  exit

interface Se2/0
  ip address 10.0.12.1 255.255.255.252
  clock rate 64000
  no shutdown
  exit

! -- OSPF --
router ospf 1
  network 192.168.1.0 0.0.0.255 area 0
  network 10.0.12.0 0.0.0.3 area 0
  exit

! -- DHCP --
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp pool LAN1
  network 192.168.1.0 255.255.255.0
  default-router 192.168.1.1
  dns-server 8.8.8.8
  exit

end
copy running-config startup-config
```

R2 with OSPF + DHCP

```
enable
configure terminal
hostname R2

interface Se3/0
  ip address 10.0.12.2 255.255.255.252
  no shutdown
  exit

interface Fa0/0
  ip address 192.168.2.1 255.255.255.0
  no shutdown
  exit

router ospf 1
  network 192.168.2.0 0.0.0.255 area 0
  network 10.0.12.0 0.0.0.3 area 0
  exit

ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp pool LAN2
  network 192.168.2.0 255.255.255.0
  default-router 192.168.2.1
  dns-server 8.8.8.8
  exit

end
copy running-config startup-config
```

10. Verification Commands Master List

Command	What It Shows
<code>show ip interface brief</code>	All interfaces, IPs, and up/down status
<code>show ip route</code>	Routing table (C=connected, R=RIP, O=OSPF, S=static)
<code>show running-config</code>	Current live config
<code>show ip protocols</code>	Which routing protocol is active
<code>show ip ospf neighbor</code>	OSPF neighbor adjacencies
<code>show ip rip database</code>	RIP learned routes
<code>show ip dhcp binding</code>	DHCP leases given out
<code>show ip nat translations</code>	Active NAT mappings
<code>show controllers serial X/X</code>	Check DCE/DTE on serial port
<code>ping X.X.X.X</code>	Test connectivity
<code>traceroute X.X.X.X</code>	Path packets take

11. Troubleshooting Checklist (When Ping Fails)

If your ping fails, check these in order:

1. **Is the interface UP?** → `show ip interface brief`
Look for “up/up”. If “administratively down” → run `no shutdown`.
2. **Are IPs correct?** → Same wire = same subnet, different hosts.
3. **Did you set clock rate on DCE serial?** → `show controllers serial X/X`
4. **Is routing configured?** → `show ip route`
If destination network is missing → check your **network** commands.
5. **RIP: Did you add version 2 + no auto-summary?**
6. **OSPF: Did you use wildcard mask, not subnet mask?**
7. **NAT: Did you mark inside/outside on the correct interfaces?**
8. **DHCP: Did PCs actually get IPs?** Check PC → Desktop → IP Config.
9. **PC Gateway:** Must match the router interface IP on that subnet.

12. PC Configuration in Packet Tracer

12.1 Static IP on a PC

1. Click PC → **Desktop** → **IP Configuration**
2. Select **Static**
3. Enter:
 - **IP Address:** e.g., 192.168.1.100
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.1.1 (the router interface IP)

12.2 DHCP on a PC

1. Click PC → **Desktop** → **IP Configuration**
2. Select **DHCP**
3. IP, mask, and gateway should auto-fill within seconds.

13. Switch Configuration (If Asked)

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1

! Assign management IP (optional)
SW1(config)# interface vlan 1
```

```
SW1(config-if)# ip address 192.168.1.250 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# exit
SW1(config)# ip default-gateway 192.168.1.1
```

14. Access Control Lists (ACLs) Quick Reference

```
! Standard ACL (filters by source IP only, numbered 1-99)
R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)# access-list 1 deny any

! Apply to interface
R1(config)# interface Fa0/0
R1(config-if)# ip access-group 1 in      ! or "out"
R1(config-if)# exit

! Extended ACL (numbered 100-199, filters src+dst+port)
R1(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80
R1(config)# access-list 100 deny ip any any

! Verify
R1# show access-lists
```

15. RIP vs OSPF Quick Comparison

Feature	RIP	OSPF
Type	Distance Vector	Link State
Metric	Hop count (max 15)	Cost (bandwidth based)
Mask in config	None (classful network)	Wildcard mask required
Convergence	Slow (30s updates)	Fast
Syntax	<code>network 10.0.0.0</code>	<code>network 10.0.0.0 0.0.0.3 area 0</code>
Required extras	<code>version 2, no auto-summary</code>	Process ID, Area
Route code	R	O

16. Default Route Redistribution (If Asked)

```
! Make the border router advertise the default route to others

! -- For RIP --
R1(config)# ip route 0.0.0.0 0.0.0.0 [next-hop or exit-interface]
R1(config)# router rip
R1(config-router)# default-information originate
R1(config-router)# exit

! -- For OSPF --
R1(config)# ip route 0.0.0.0 0.0.0.0 [next-hop or exit-interface]
R1(config)# router ospf 1
R1(config-router)# default-information originate
R1(config-router)# exit
```

17. Adding WIC-2T Module in Packet Tracer (PHYSICAL SETUP)

You MUST add WIC-2T BEFORE powering on the router or connecting cables!

Default routers (e.g., 1841, 2811) only have FastEthernet ports. To get Serial ports, you need to physically install a **WIC-2T** (WAN Interface Card – 2 serial ports) module.

17.1 Step-by-Step: Adding WIC-2T

1. Click on the router in Packet Tracer
2. Go to the **Physical** tab
3. **TURN OFF the router** — click the power switch (green button on the right side of the chassis image). It must be OFF to add hardware.
4. In the left panel, find **WIC-2T** under **WAN Interface Cards**
5. **Drag and drop** WIC-2T into an empty WIC slot on the router image (the small open slot on the router chassis)
6. **Turn the router back ON** — click the power switch again
7. Wait for the router to boot (IOS loading messages)
8. Now you have **Serial0/0** and **Serial0/1** available (or **Se0/0/0**, **Se0/0/1** depending on router model)

Interface names depend on the router model:

Router 2811:	Se0/0/0, Se0/0/1 (slot/subslot/port)
Router 1841:	Se0/0/0, Se0/0/1
Router 2621:	Se0/0, Se0/1 (slot/port)
Generic Router-PT:	Se2/0, Se3/0

After adding WIC-2T, check with: **show ip interface brief** to see exact names!

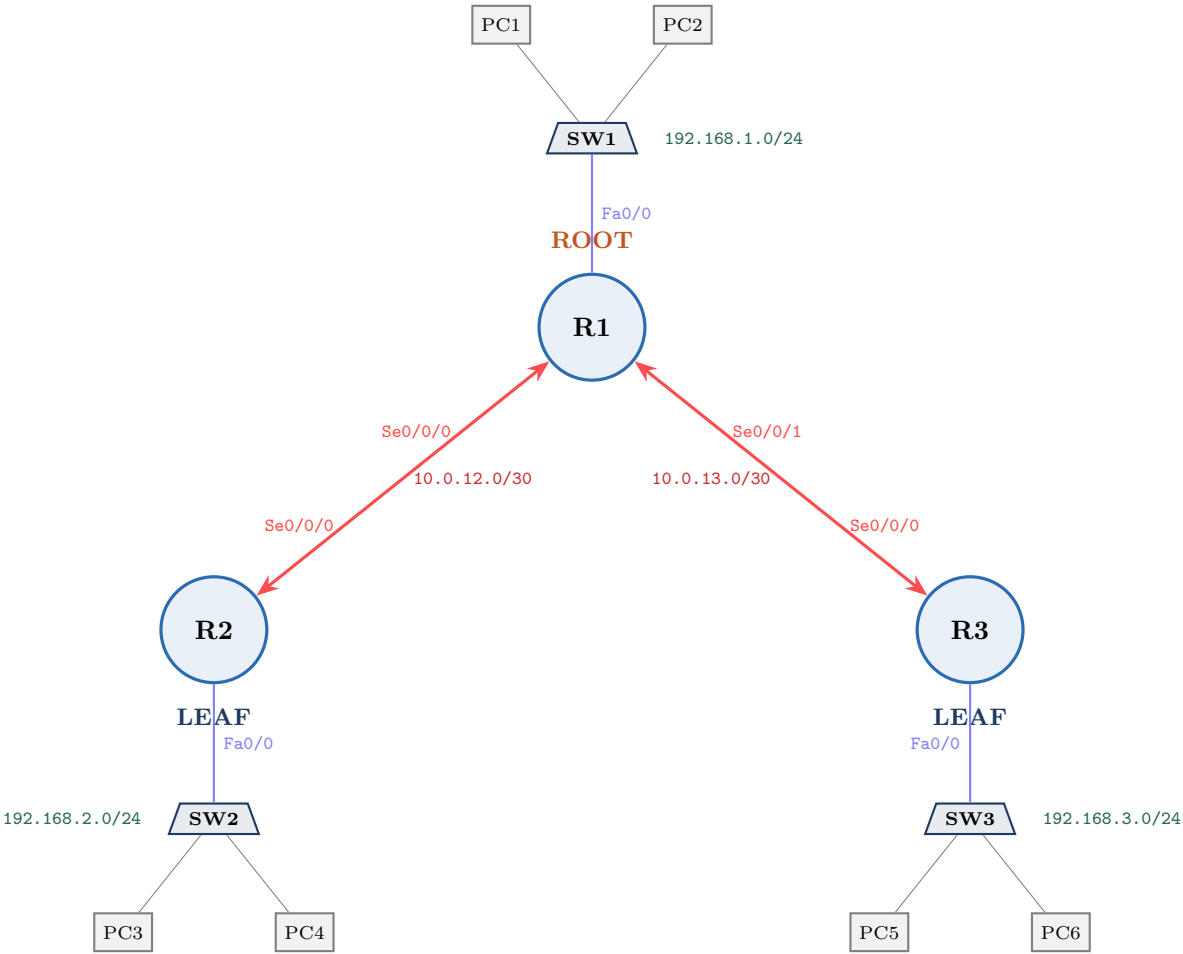
17.2 Connecting Serial Cables

1. Go to **Connections** (lightning bolt icon at bottom)
2. Choose **Serial DCE** cable (the one with the clock icon) — NOT “Serial DTE”
3. Click on Router 1 → select its serial port (e.g., Se0/0/0)
4. Click on Router 2 → select its serial port (e.g., Se0/0/0)
5. The end where you clicked **first** = **DCE side** (needs **clock rate**)
6. The end where you clicked **second** = **DTE side** (no clock rate needed)

How to remember: The router you click **FIRST** gets the clock icon = DCE = needs **clock rate**.
If unsure, run **show controllers serial 0/0/0** on both routers and look for “DCE” or “DTE”.

18. FULL EXAMPLE: 3 Routers, Hub-and-Spoke, All Serial, RIP

18.1 Topology: R1 (root) connected to R2 and R3 (leaves)



18.2 IP Address Plan

Device	Interface	IP Address	Description
R1 (Root)	Fa0/0	192.168.1.1 / 255.255.255.0	LAN (to SW1, PCs)
	Se0/0/0	10.0.12.1 / 255.255.255.252	Serial to R2 (DCE)
	Se0/0/1	10.0.13.1 / 255.255.255.252	Serial to R3 (DCE)
R2 (Leaf)	Fa0/0	192.168.2.1 / 255.255.255.0	LAN (to SW2, PCs)
	Se0/0/0	10.0.12.2 / 255.255.255.252	Serial to R1 (DTE)
R3 (Leaf)	Fa0/0	192.168.3.1 / 255.255.255.0	LAN (to SW3, PCs)
	Se0/0/0	10.0.13.2 / 255.255.255.252	Serial to R1 (DTE)
PC1, PC2	Fa0	DHCP or 192.168.1.x/24	Gateway: 192.168.1.1
PC3, PC4	Fa0	DHCP or 192.168.2.x/24	Gateway: 192.168.2.1
PC5, PC6	Fa0	DHCP or 192.168.3.x/24	Gateway: 192.168.3.1

Hub-and-spoke: R1 is the “hub” (root). It has 2 serial interfaces, one going to each leaf. R2 and R3 are “spokes” (leaves). They each have only 1 serial interface going up to R1. R1 needs WIC-2T (gives 2 serial ports). R2 and R3 each need WIC-2T (use 1 of the 2 ports).

18.3 Complete Config: R1 (Root — Hub)

R1 — ROOT router (HUB) — 2 serial links
RIP — OSPF — DHCP — NAT — Subnetting

```
enable
configure terminal
hostname R1

! ---- LAN interface (to switch/PCs) ----
interface FastEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 exit

! ---- Serial to R2 (DCE side --- needs clock rate) ----
interface Serial 0/0/0
 ip address 10.0.12.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit

! ---- Serial to R3 (DCE side --- needs clock rate) ----
interface Serial 0/0/1
 ip address 10.0.13.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit

! ---- RIP ----
router rip
 version 2
 no auto-summary
 network 192.168.1.0
 network 10.0.12.0
 network 10.0.13.0
 exit

end
copy running-config startup-config
```

18.4 Complete Config: R2 (Leaf)

R2 — LEAF router — 1 serial link to R1

```
enable
configure terminal
hostname R2

! ---- LAN interface ----
interface FastEthernet 0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
 exit

! ---- Serial to R1 (DTE side --- NO clock rate) ----
interface Serial 0/0/0
 ip address 10.0.12.2 255.255.255.252
 no shutdown
 exit

! ---- RIP ----
router rip
 version 2
 no auto-summary
 network 192.168.2.0
 network 10.0.12.0
 exit

end
copy running-config startup-config
```

18.5 Complete Config: R3 (Leaf)

R3 — LEAF router — 1 serial link to R1

```
enable
configure terminal
hostname R3

! ---- LAN interface ----
interface FastEthernet 0/0
 ip address 192.168.3.1 255.255.255.0
 no shutdown
 exit

! ---- Serial to R1 (DTE side --- NO clock rate) ----
interface Serial 0/0/0
 ip address 10.0.13.2 255.255.255.252
 no shutdown
 exit

! ---- RIP ----
router rip
 version 2
 no auto-summary
 network 192.168.3.0
 network 10.0.13.0
 exit

end
copy running-config startup-config
```

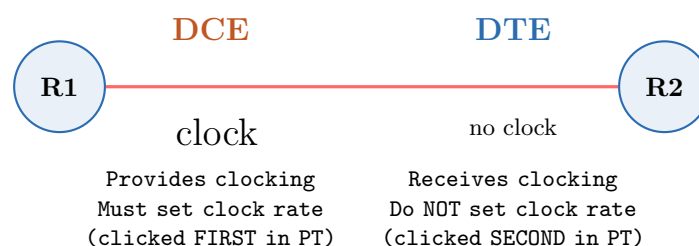
18.6 Verification After Setup

```
! On R1 (root):
R1# show ip interface brief
! Should see: Fa0/0 up/up, Se0/0/0 up/up, Se0/0/1 up/up

R1# show ip route
! Should see:
! C 192.168.1.0/24 (directly connected Fa0/0)
! C 10.0.12.0/30 (directly connected Se0/0/0)
! C 10.0.13.0/30 (directly connected Se0/0/1)
! R 192.168.2.0/24 via 10.0.12.2 (learned from R2 via RIP)
! R 192.168.3.0/24 via 10.0.13.2 (learned from R3 via RIP)

! Test: From PC1 (192.168.1.x) ping PC3 (192.168.2.x) -- should work!
! Test: From PC1 ping PC5 (192.168.3.x) -- should work!
! Test: From PC3 ping PC5 -- should work (goes through R1)!
```

19. Clock Rate & DCE/DTE Deep Dive



What is clock rate?

Serial links need a clock signal to synchronize data transmission. In real life, the ISP provides this. In Packet Tracer, ONE side of the cable must provide it.

Rules:

1. Only the **DCE** side sets clock rate
2. If you don't set it, the serial link stays **DOWN** (won't come up even with no shutdown)
3. To check which side you are: `show controllers serial 0/0/0`
4. Look for the line that says "**DCE V.35**" or "**DTE V.35**"

How to check and set clock rate

```
! CHECK which end you are:
R1# show controllers serial 0/0/0
! Look for: "DCE V.35" or "DTE V.35"

! If DCE, set clock rate:
R1(config)# interface Serial 0/0/0
R1(config-if)# clock rate 64000
R1(config-if)# exit

! If DTE, do NOTHING (no clock rate command needed).
! Common clock rate values: 64000, 128000, 256000, 512000, 1000000, 2000000
```

Symptom if you forget clock rate:

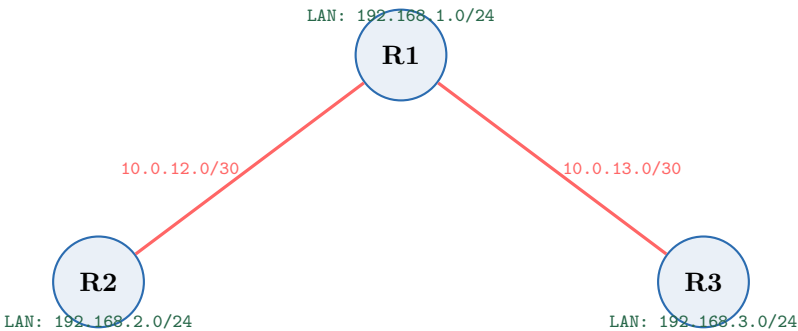
`show ip interface brief` will show the serial interface as:

Serial0/0/0	10.0.12.1	YES	up	down
-------------	-----------	-----	----	------

Status = up but Protocol = **down** means the link layer can't sync → missing clock rate on DCE side!

20. Static Routing for 3 Routers (Hub-and-Spoke)

If the question says “use static routing” instead of RIP/OSPF, you manually tell each router how to reach every network it doesn't directly connect to.



20.1 What Routes Does Each Router Need?

Router	Knows directly	Needs static routes to
R1	192.168.1.0, 10.0.12.0, 10.0.13.0	192.168.2.0, 192.168.3.0
R2	192.168.2.0, 10.0.12.0	192.168.1.0, 192.168.3.0, 10.0.13.0
R3	192.168.3.0, 10.0.13.0	192.168.1.0, 192.168.2.0, 10.0.12.0

20.2 Static Route Commands

R1 (Root) — Static Routes

RIP — OSPF — DHCP — NAT — Subnetting

```
! R1 already knows 192.168.1.0, 10.0.12.0, 10.0.13.0
! Needs routes to R2's LAN and R3's LAN

R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.12.2
! "To reach 192.168.2.0, send packets to 10.0.12.2 (R2)"

R1(config)# ip route 192.168.3.0 255.255.255.0 10.0.13.2
! "To reach 192.168.3.0, send packets to 10.0.13.2 (R3)"
```

R2 (Leaf) — Static Routes

```
! R2 already knows 192.168.2.0, 10.0.12.0
! Needs routes to R1's LAN, R3's LAN, and R1-R3 serial link

R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.12.1
R2(config)# ip route 192.168.3.0 255.255.255.0 10.0.12.1
R2(config)# ip route 10.0.13.0 255.255.255.252 10.0.12.1
! All traffic goes via R1 (10.0.12.1), since R2 only connects to R1

! OR use a default route (simpler for leaf nodes):
R2(config)# ip route 0.0.0.0 0.0.0.0 10.0.12.1
! "Send everything I don't know about to R1"
```

R3 (Leaf) — Static Routes

```
! Same logic: everything goes via R1

R3(config)# ip route 192.168.1.0 255.255.255.0 10.0.13.1
R3(config)# ip route 192.168.2.0 255.255.255.0 10.0.13.1
R3(config)# ip route 10.0.12.0 255.255.255.252 10.0.13.1

! OR use a default route:
R3(config)# ip route 0.0.0.0 0.0.0.0 10.0.13.1
```

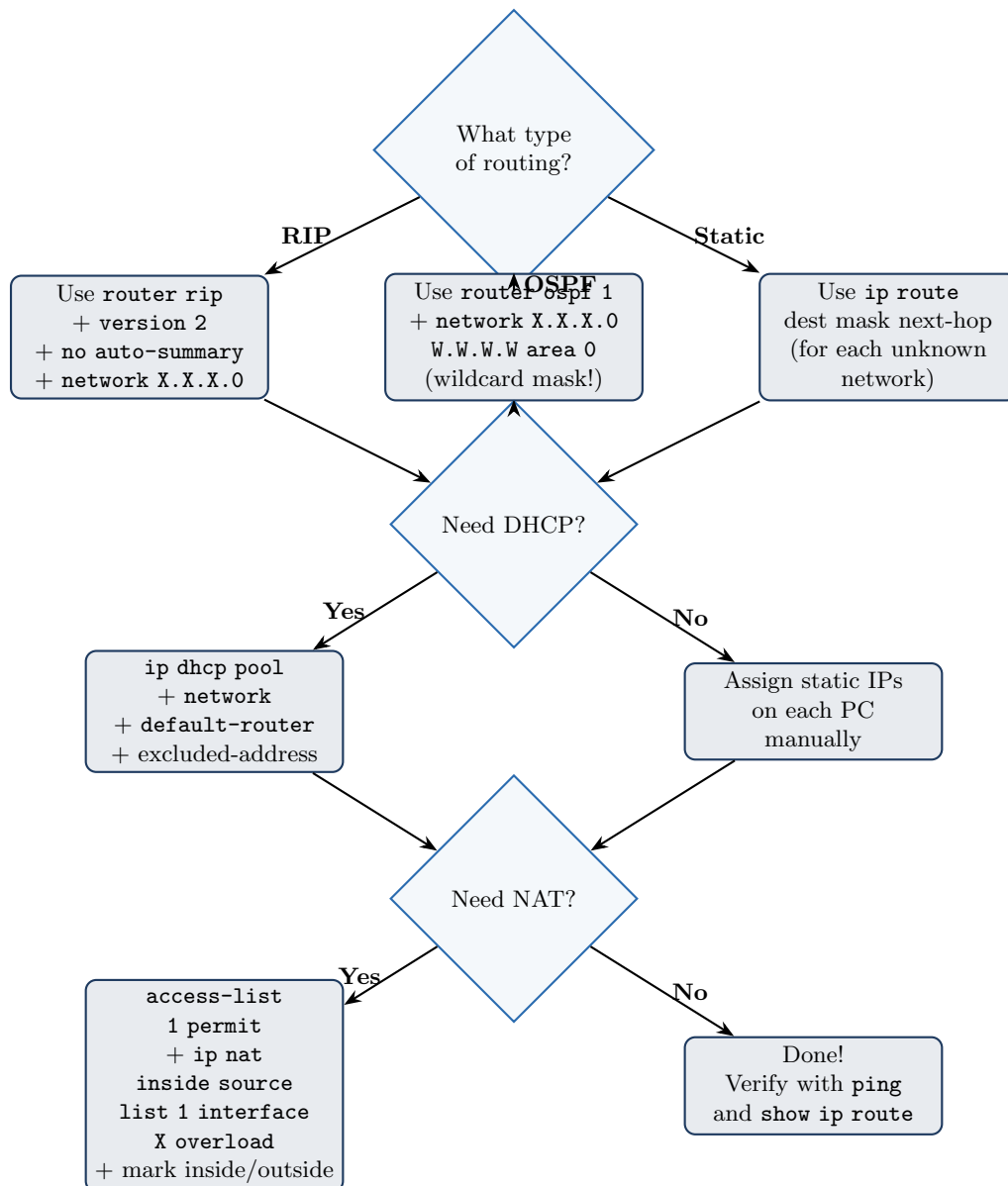
Shortcut for leaf routers: Since all traffic from a leaf must go through R1 (the hub), you can replace all the individual static routes with a single **default route**:

```
ip route 0.0.0.0 0.0.0.0 [next-hop-to-R1]
```

This says “for any destination I don’t know, send it to R1.”

This is much simpler and achieves the same result!

21. Quick Decision Flowchart



Good luck on your eval!

**Remember: no shutdown, correct IPs, and save your config.
When in doubt, show ip interface brief and show ip route.**